

ABU SALEH MOSA FAISAL

Bioinformatics Analyst

Markey Cancer Center, University of Kentucky

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Education

College of Public Health, University of Kentucky

PhD in Epidemiology and Biostatistics

August 2025- Present

Lexington, Kentucky

Gatton College of Business and Economics, University of Kentucky

Master of Business Administration (MBA)

August 2023- May 2026

Lexington, Kentucky

New Mexico Institute of Mining and Technology

MS in Mathematics with specialization OR and Statistics

December 2021

Socorro, New Mexico

University Of Dhaka

BSc. in Applied Statistics

March 2013

Dhaka, Bangladesh

Academic Awards and Honors

- Awarded with scholarship to fund MS in Mathematics by New Mexico Institute of Mining and Technology, USA (2018).

Technical Skills

Programming & Statistical Computing: R, Python, SAS, SQL, Java, C, C++

Bioinformatics & Data Science: Bioconductor, Seurat, DESeq2, edgeR, limma, GSEA, TCGAblinks, maftools, CIBERSORTx

Databases & Computing: PostgreSQL, MySQL, NoSQL, Linux/Unix, Docker, Git

Visualization & Reporting: Tableau, Power BI, Microsoft Excel, IPA, R Markdown, Quarto, L^AT_EX

Professional Experience

Markey Cancer Center, University of Kentucky

Bioinformatics Analyst, PI: Chi Wang, Ph.D.

March 2023 – Present

Lexington, Kentucky

- Provide comprehensive biostatistical, bioinformatics, and data science consultation to basic scientists, translational researchers, and clinical investigators, supporting projects from study design through data analysis, interpretation, and publication.
- Collaborate with multidisciplinary research teams on study implementation and management, experimental design, statistical analysis plans, power calculations, and reproducible analytical workflows.
- Perform advanced analyses of high-throughput multi-omics datasets, including bulk RNA-seq, scRNA-seq, spatial transcriptomics, ATAC-seq, scATAC-seq, ChIP-seq, proteomics, metabolomics, microbiome, and microarray data.
- Conduct genomic data mining using large public repositories including TCGA, GDC, cBioPortal, GEO, SRA, GTEx, Single Cell Portal, UCSC Xena, and other publicly available genomic resources.
- Develop end-to-end computational pipelines for quality control, preprocessing, normalization, differential expression analysis, pathway enrichment, functional annotation, and biological interpretation.
- Analyze genomic alterations including somatic mutation profiling, copy number variation (CNV), structural variants, mutational signatures, and genomic instability.
- Perform survival analysis, prognostic modeling, biomarker discovery, clinical outcome analysis, and clinical trial data analysis using advanced statistical methodologies.
- Apply statistical machine learning, predictive modeling, clustering, dimensionality reduction, feature selection, and classification methods to identify clinically relevant molecular signatures.
- Integrate diverse multi-omics datasets to investigate complex biological mechanisms and identify therapeutic targets and biomarkers.
- Provide statistical consultation for experimental design, hypothesis testing, longitudinal studies, generalized linear models, mixed-effects models, Bayesian analysis, and causal inference.
- Support investigators in preparing competitive grant applications, statistical sections of proposals, manuscripts, figures, supplementary analyses, and responses to peer reviewers.
- Develop reproducible computational workflows and perform statistical programming using R, Python, SAS, Bash, SQL, and Linux/Unix environments while following best practices in reproducible research.

- Utilize leading bioinformatics and statistical toolkits including Seurat, Bioconductor, DESeq2, edgeR, limma, Harmony, Signac, Monocle, CellChat, inferCNV, maftools, GSEA, clusterProfiler, ComplexHeatmap, TCGAbiolinks, CIBERSORTx, WGCNA, Scikit-learn, TensorFlow, PyTorch, Snakemake, Nextflow, Docker, and Git.
- Communicate complex analytical findings effectively through scientific reports, presentations, publications, and collaborative discussions with investigators across multiple biomedical disciplines.

LSU Health Science Center

May 2022 – March 2023

Biostatistician, PI: Mohammad Alfred Nobel Bhuiyan, Ph. D.

Shreveport, Louisiana

- Designed and executed advanced statistical analyses for clinical trials, epidemiologic studies, and translational biomedical research, ensuring methodological rigor and reproducibility.
- Developed scalable analytical pipelines using **R**, **Python**, and **SAS** for the analysis of complex clinical, genomic, and population-level datasets.
- Served as the statistical collaborator for peer-reviewed publications, grant proposals, conference abstracts, and scientific presentations by developing statistical methodologies and interpreting analytical results.
- Provided expert statistical consultation on study design, power and sample size calculations, data management, analytical strategies, and interpretation of findings to multidisciplinary research teams.
- Directed comprehensive analyses of the **HCUP National Inpatient Sample (NIS)** database to characterize substance use-related disease burden, healthcare utilization, and clinical outcomes at the national level.

Research Experience

New Mexico Institute of Mining and Technology

January 2018 – May 2021

Graduate Researcher, PI: Oleg Makhnin, Ph.D.

Socorro, New Mexico

- Developed **gibbSeq2**, an innovative Bayesian statistical framework for RNA-seq gene set enrichment analysis, providing robust probabilistic inference for pathway discovery.
- Developed predictive machine learning models integrating clinical, genomic, and multi-omics data to identify biomarkers associated with cancer treatment outcomes.
- Applied advanced statistical modeling to evaluate the relationship between serum creatinine concentrations and phenacetin exposure in women, generating evidence for epidemiological and clinical research.

University of Dhaka

Jan. 2013 – Dec. 2013

Research Assistant, Supervisor: M. Shafiqur Rahman, Ph. D.

Dhaka, Bangladesh

- Applied advanced statistical and epidemiological methods to identify predictors of low birth weight among Bangladeshi newborns, generating insights to inform maternal and neonatal health interventions.

Teaching Experience

New Mexico Institute of Mining and Technology

August 2018 – December 2021

Graduate Teaching Assistant

Socorro, New Mexico

- Delivered laboratory and recitation instruction for undergraduate **Applied Statistics** and **Calculus I & II** courses, fostering quantitative reasoning and problem-solving skills through active learning techniques.
- Provided individualized academic mentoring during office hours, assisting students with mathematical concepts, statistical methods, homework, and exam preparation.
- Evaluated assignments, laboratory reports, quizzes, and examinations, delivering constructive feedback to enhance student learning outcomes.
- Supported faculty in curriculum implementation, classroom management, assessment activities, and instructional material development while ensuring compliance with departmental academic standards.
- Maintained a collaborative and inclusive learning environment that encouraged student engagement, critical thinking, and academic success.

Publications

- Xing, Diensn G., Farhan Mohiuddin, Md Shenuarin Bhuiyan, Md Ismail Hossain, Zaki Al-Yafeai, Abu Saleh Mosa Faisal, Nicholas E. Goeders et al. "Prevalence and patterns of methamphetamine use and mental health disparity in the United States." *Nature Mental Health* (2024): 1-9.
- Khan, Muhammad Z., Rene Alvarez, Mohammad Alfrad Nobel Bhuiyan, Abu Saleh Mosa Faisal, Parker O'Neill, Muhammad Siddiqui, Praneet Kaki et al. "Clinical Outcomes of Transcatheter Aortic Valve Replacement (TAVR) vs Surgical Aortic Valve Replacement (SAVR) in Patients With Sarcoidosis." *Cureus* 16, no. 6 (2024).
- Rives, Taylor A., Abu Saleh Mosa Faisal, Jinpeng Liu, Hye Sook Chon, Casey Cosgrove, Bradley Corr, Robert Dood et al. "Comparison of the clinical and genomic profiles of endometrial cancer in Appalachian and non-Appalachian patients." (2024): 5591-5591.

- Pruitt, Aaron, Jinpeng Liu, Abu Saleh Mosa Faisal, Feitong Lei, Bin Huang, Eric B. Durbin, Jong Cheol Jeong et al. "Young lung adenocarcinoma: Molecular analysis pointing to early acquisition of aggressive mutations." (2024): e20089-e20089.
- Geisen, Mariah, Josiane weber-Tessmann, Courtney Kelson, Daheng He, Chi Wang, Abu Saleh Mosa Faisal, Jill Kolesar, and Yekaterina Zaytseva. "Inhibition of FASN postpones development of resistance to BRAF inhibitors in colorectal cancer." *Cancer Research* 84, no. 6, Supplement (2024): 4760-4760.
- Al-Yafeai, Zaki, Shafaqat Ali, Jimmy Brown, Maamannan Venkataraj, Md Shenuarin Bhuiyan, Abu Saleh Mosa Faisal, Kenneth Densmore et al. "Cardiomyopathy-associated hospital admissions among methamphetamine users: geographical and social disparities." *JACC: Advances* (2024): 100840.
- Mubashir, Maryam, Victoria Andrus, David Okuampa, Michelle Neice, Elizabeth Armstrong, Hailey Canezaro, Ross Dies et al. "Short-term Outcome of Peroral Endoscopic Myotomy Performed by the Same Endoscopist on Achalasia and Nonachalasia Esophageal Motility Disorders." *Surgical Laparoscopy Endoscopy & Percutaneous Techniques* 33, no. 6 (2023): 577-582.
- Xing, Diensn G., Teresa Horan, Md Shenuarin Bhuiyan, Abu Saleh Mosa Faisal, Kenneth Densmore, Kevin S. Murnane, Nicholas E. Goeders et al. "Social-geographic disparities in suicidal ideations among methamphetamine users in the USA." *Psychiatry research* 329 (2023): 115524.
- Raza, Syed Musa, Daniyal Raza, Elizabeth Armstrong, Victoria Andrus, Grace Ann McCurdy, Francesca Weis, Michelle Neice,..., Abu Saleh Mosa Faisal et al. "SHORT-TERM OUTCOMES OF GERD IN GASTROPARESIS PATIENTS AFTER GASTRIC PER ORAL ENDOSCOPIC PYLOROMYOTOMY-FINAL RESULTS." *Gastrointestinal Endoscopy* 97, no. 6 (2023): AB753.
- Khan, Jahidur Rahman, Md Mazharul Islam, Abu Saleh Mosa Faisal, Humayera Islam, and K. Shuvo Bakar. "Quantification of urbanization using night-time light intensity in relation to women's overnutrition in Bangladesh." *Journal of Urban Health* 100, no. 3 (2023): 562-571.
- Bhuiyan, Md Shenuarin, Abu Saleh Mosa Faisal, Maamannan Venkataraj, Nicholas E. Goeders, Steven R. Bailey, Steven A. Conrad, John A. Vanchiere, A. Wayne Orr, Christopher G. Kevil, and Mohammad Alfrad Nobel Bhuiyan. "Disparities in Prevalence and Trend of Methamphetamine-Associated Cardiomyopathy in the United States." *Journal of the American College of Cardiology* 81, no. 18 (2023): 1881-1883.
- Mubashir, Maryam, Syed Musa Raza, Shazia Rashid, David Okuampa, Daniyal Raza, Allison Derise, Kwabena O. Asafo-Agyei, Abu Saleh M. Faisal, Jehan Abdulsattar, and Qiang Cai. "Sa1473 POST-CHOLECYSTECTOMY SYNDROME; REVALUATING AND REDEFINING." *Gastroenterology* 164, no. 6 (2023): S-398.
- Hafiz, Nazar, Aditya Vyas, Omar Khan, Janak Bahirwani, Syed Musa Raza, Abu Saleh M. Faisal, Mohammad Alfrad Nobel Bhuiyan et al. "Su1647 ANALYSIS OF EACH COMPONENT OF GASTROPARESIS CARDINAL SYMPTOM INDEX (GCSI) IN PREDICTING THE RISK OF GASTROPARESIS IN PATIENT POPULATION WITHOUT THE DIAGNOSIS OF GASTROPARESIS." *Gastroenterology* 164, no. 6 (2023): S-635.
- Khan, Jahidur Rahman, Abu Saleh Mosa Faisal, Sukanta Das, and Nabil Awan. "Identifying overnutrition risk groups in Bangladeshi married women based on sociodemographic factors: A classification and regression tree model." *Obesity Medicine* 33 (2022): 100425.
- Faisal, Abu Saleh Mosa. *gibbseq2: A Bayesian Model to Detect Differential Expression of Gene Sets in RNA-seq Data*. New Mexico Institute of Mining and Technology, 2021.

Conference Presentation

- "Comparison of the Clinical and Genomic Profiles of Endometrial Cancer in Appalachian and Non-Appalachian Patients". 2024 ORIEN Scientific Summit. Tampa, FL, USA
- "Comparing relative abundance of microbial families during bio-remediation, using High throughput data and mixed models". Integrative Omics Conference-2019, Santa Fe, NM, USA
- "Profile analysis of microbial community using one-way anova model". NMT Student Research Symposium-2019, Socorro, NM, USA

Leadership & Service

- General Secretary of Bangladesh Student Association at University of Kentucky.
- Volunteer for Registration Responsibilities in “International Conference on Applied Statistics-2014” (ICAS-2014).
- Member of Tech Policy Group at New Mexico Tech

Workshops & Training

- NCGR 2022 virtual Bioinformatics Workshop on Differential gene Expression
- NCGR Fall 2021 Virtual Bioinformatics Workshop on Metagenomics and Metatranscriptomics